

Yen-Hsing (Julius) Lin

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Publication list: [ADS](#)

Personal Website (more detailed information!)



ACADEMIC EXPERIENCE

Institute of Astronomy and Astrophysics, Academia Sinica

Taipei, Taiwan

Undergraduate Student, PI: Dr. Hiroyuki Hirashita

2019 – 2021

- Investigating the effect of geometry and dust grain size distribution on the SED of high redshift galaxies using radiation transfer code SKIRT.

Institute of Astronomy, NTHU

Hsinchu, Taiwan

Undergraduate / Master Student, PI: Prof. Hsiang-Yi Karen Yang

2020 – 2023

- Investigating cosmic ray dominated AGN jet mode feedback with 3D CRMHD simulations using FLASH, with focus on cosmic ray composition and odd radio circles (ORCs).

Department of Astronomy, UCSD

California, US

PhD Student, PI: Dusan Keres (FIRE collaboration)

2024 – Current

- Constrains on cosmic ray diffusion coefficient using $L_{\text{IR}} - L_{\gamma}$ relation.
- Understand the origin of dust in the circumgalactic medium (CGM).

SCIENTIFIC PUBLICATIONS

1. **Yen-Hsing Lin**, Hiroyuki Hirashita, Peter Camps, Maarten Baes, "Geometry effects on dust attenuation curves with different grain sources at high redshift", Monthly Notices of the Royal Astronomical Society, Volume 507, Issue 2, October 2021, Pages 2755-2765
2. Yuan-Ming Hsu, Hiroyuki Hirashita, **Yen-Hsing Lin**, Peter Camps, Maarten Baes, "Effects of dust sources on dust attenuation properties in IllustrisTNG galaxies at $z \sim 7$ ", Monthly Notices of the Royal Astronomical Society, Volume 519, Issue 2, February 2023, Pages 2475-2485.
3. **Yen-Hsing Lin**, H.-Y. Karen Yang, Ellis R. Owen, "Evolution and feedback of AGN Jets of different Cosmic-ray Composition", Monthly Notices of the Royal Astronomical Society, Volume 520, Issue 1, March 2023, Pages 963-975.
4. **Yen-Hsing Lin**, H.-Y. Karen Yang, "AGN jet-inflated bubbles as possible origin of odd radio circles", ApJ, Vol 974, Page 269.
5. Kokori, ..., **Yen-Hsing Lin**, "ExoClock Project. IV. A Homogeneous Catalog of 620 Updated Exoplanet Ephemerides", ApJS, Vol 283, Issue 1, id.5, 27 pp.
6. Isabel S. Sands, Philip F. Hopkins, Sam B. Ponnada, Lina Necib, Dušan Kereš, **Yen-Hsing Lin**, "Galactic Center Gamma-Ray Emission in MHD Galaxy Formation Simulations with Full Cosmic Ray Spectra", submitted
7. **Yen-Hsing Lin**, Dusan Keres, " $L_{\text{IR}} - L_{\gamma}$ relation of FIRE galaxies", in prep

AWARDS AND SCHOLARSHIPS

NSTC: National Science and Technology Council; **NCTS**: National Center for Theoretical Sciences

TPS: Physical Society of Taiwan; **ASROC**: The Astronomical Society of the Republic of China (Taiwan)

- **2024 J.Yang Scholarship**
 - UCSD, 2024.05.20.
- **2024 NTHU Hsing Jian award**
 - The award recognize for outstanding performance in extracurricular activities, 2024.04.11.
- **UCSD A&A Inclusive Excellence Award**
 - UCSD, 2024.04.10.
- **The Katzin Fellowship**
 - UCSD, 2024.04.10.
- **2024 TPS annual meeting oral competition: Merit award**
 - Title: Odd radio circles as cosmic-ray dominated AGN jet-inflated bubbles, 2024.01.26.
- **2023 TPS Postgraduates Student Thesis Award (Special Excellence)**
 - Master thesis: Simulating the effects of cosmic-ray dominated black-hole jets in galaxy clusters, 2023.12.18.
- **2023 NSTC Graduate Research Fellowship**
 - 2023.10.19. 4 years. 40000 NTD (~ 1300 USD) per month.
- **2023 ASROC oral competition: Merit award**
 - Title: Could odd radio circles be end-on AGN jet inflated bubbles?, 2023.05.21
- **2023 NTHU College of Science Elite Student Award**
- **2023 TPS annual meeting poster competition: Merit award**
 - Title: Simulating the effects of cosmic-ray dominated jets in cool-core clusters, 2023.01.18
- **2022 TPS Undergraduates Student Thesis Award**
 - Paper: “Geometry effects on dust attenuation curves with different grain sources at high redshift”, 2022.12.12.
- **2022 NCTS Student Outstanding Paper Award**
 - Paper: “Geometry effects on dust attenuation curves with different grain sources at high redshift”, 2022 Summer.
- **2022 TPS Annual Meeting Poster Competition: Honorable Mention**
 - Annual Meeting of the Physical Society of Taiwan, 2022.01.25.
- **2021 NTHU Physics Poster Competition: 1st place**
 - NTHU Physics Department. 2021.12.29.

INVITED TALKS

1. **NTHU IoA Special Lunch Seminar**
 - Cosmic Dust beyond Galaxies: Progress and Open Question in the Circumgalactic Medium. 2026.07.29
2. **ASIAA lunch talk**
 - $L_{\text{IR}} - L_{\gamma}$ relation of FIRE-2 galaxies. 2026.07.20
3. **U. Arizona TACO**
 - $L_{\text{IR}} - L_{\gamma}$ relation of FIRE-2 galaxies. 2026.05.27
4. **NTHU IoA Colloquium**
 - $L_{\text{IR}} - L_{\gamma}$ relation of FIRE-2 galaxies. 2025.12.19
5. **NTHU IoA Seminar Series**
 - AGN jet-inflated bubbles as possible origin of odd radio circles. 2024.05.21
6. **NCTS annual theory meeting**
 - Simulating the effects of cosmic-ray dominated jets in galaxy clusters. 2022.12.14

7. NCTS outstanding student paper award ceremony

- Geometry effect on dust attenuation curves of different grain sources in high redshift, 2022.09.23

8. NCTS astrophysics seminar

- Evolution of AGN Jets of different cosmic-ray composition, 2022.08.26.

9. NTU physics department student seminar

- Geometry effect on dust attenuation in high redshift / Evolution of AGN jets of different cosmic-ray composition, 2021.05.13

OBSERVATION PROPOSALS

1. Lulin Observatory | 2024B, 2024C, 2025A, 2025B, 2025C

- PI. Title: Exoplanet ephemerides refinement for the Ariel Space Telescope. ~ 100 hours awarded in total.

2. Steward Observatory Kuiper 61 inches telescope | 2025A

- Co-I. Title: Ephemerides refinement of sub-Neptune TOI-4479b. Allotted 2 nights.

3. Giant Metrewave Radio Telescope (uGMRT) | 2024, 2025

- Co-I. Title: Low frequency observations of the odd radio circle ORC J0102-2450. Proposal ID: 48_18, 47_052. Allotted 56 hours.

4. MeerKAT | 2024

- Co-I. Title: Searching for the origin of the Odd Radio Circles. Proposal ID: MKT-242432. Allotted 27 hours.

5. XMM-Newton X-ray telescope | 2024

- Co-I. Title: Searching for the origin of the odd radio circles (ORCs). Proposal ID: 096411. Allotted 20 ks.

RECENT CONTRIBUTED TALKS

Since 2023.

1. Cosmic Ray Transport Workshop 2026 (Oral)

- $L_{\text{IR}} - L_{\gamma}$ relation of FIRE galaxies, 2026.05.18

2. GalFresca 2025 (Oral)

- $L_{\text{IR}} - L_{\gamma}$ relation of FIRE galaxies, 2025.09.16

3. 2024 UCSD Astro Journal Club (Oral)

- How I became an amateur planet hunter, 2025.02.21

4. 2024 UCSD Astro Journal Club (Oral)

- Could Odd Radio Circles be end-on AGN jet inflated bubbles?, 2024.10.18

5. 2024 TPS annual meeting (Oral)

- Could Odd Radio Circles be end-on AGN jet inflated bubbles?, 2024.01.25

6. 2024 Taipei Astronomy Workshop (Oral)

- Could Odd Radio Circles be end-on AGN jet inflated bubbles?, 2024.01.09

7. 2023 ASROC annual meeting (Oral)

- Could Odd Radio Circles be end-on AGN jet inflated bubbles?, 2023.05.21

8. 2023 TPS annual meeting (Poster)

- Simulating the effects of cosmic-ray dominated jets in cool-core clusters, 2023.01.18

This CV is created by modifying Doruk Çetin's Simple-CV template.

Last update on August 3, 2026.